

Protocol

Neurosurgical prophylaxis

1 Scope

For use within neurosurgery.

2 Purpose

To ensure that appropriate antibiotics are used routinely in neurosurgical procedures.

3 Introduction

Antibiotic prophylaxis before neurosurgery can be a useful adjunct to meticulous aseptic technique. Prophylactic antibiotics administered before craniotomy reduced rates of postoperative meningitis by approximately one-half in a recent meta-analysis.

In other studies of prophylaxis before craniotomy and shunt surgery it achieves a significant reduction (~50%) in infection rates. However, baseline infection rates are low (usually 2-6%), hence the number needed to treat (NNT) to prevent one neurosurgical infection is of the order of 14-16 patients. The pathogens recovered are *Staph aureus* (52%), coagulase-negative staphylococci (15%), streptococci (9%), Gram-negative bacilli (16%), others (8%).

The great majority of organisms infecting surgical wounds in this hospital are susceptible to flucloxacillin and gentamicin; these remain narrow spectrum antibiotics and are recommended in the interest of antimicrobial stewardship.

The timing of the prophylactic dose is critical; NICE recommend that antibiotic prophylaxis is given at the beginning of anaesthesia. Infection rate increases significant when prophylaxis is given after or more than two hours before the surgical incision.

The use of additional doses of antibiotics is widely recommended including by National Institute for Health and Care Excellence (NICE) when the duration of the operation is longer than the half-life of the antibiotic prescribed. Recommended doses and dosing intervals are given in the section named [repeated doses post operatively and during long operations](#).

A considerable literature review shows no significant benefit for post-operative doses in general and orthopaedic implant surgery, and by implication,

neurosurgery. Antibiotic prophylaxis has been shown to have no benefit in cerebrospinal fluid (CSF) fistula or leaks.

This guideline has been updated to optimise flucloxacillin (dose increased to 2g in adults) and gentamicin (dose increased to 3mg/kg in adults) dosing for neurosurgical prophylaxis. Repeat doses of gentamicin are not recommended for long operations.

4 Undertaken by

This protocol is for the doctors, nurses and pharmacists involved in the management of these patients.

5 Recommended practice

5.1 Clean implant and non-implant craniofacial and spinal procedures – surgery through healthy skin (see note 1)

Non – implant procedures include:

- Percutaneous radiofrequency ablation for trigeminal neuralgia
- Non-implant spinal surgery

Implant procedures include:

- Insertion of CSF shunts: (see notes 2 & 3)
- Insertion of bone flaps: acrylic/ titanium/ bone
- External CSF drains (see note 4)
- Deep brain stimulators
- Spinal cord stimulators
- Spinal implant surgery (see note 5 and section 5.7.2)

First choice:

Adults:

**Single dose flucloxacillin 2 g IV
plus gentamicin 3 mg/kg IV**

Paediatrics:

One month–18 years

**Single dose flucloxacillin 50 mg/kg IV (max. 2 g)
plus gentamicin IV at induction (for dosage see below)**

Gentamicin dosing:

One month-12 years

2.5mg/kg IV (max 120mg)

Child 12-18 years

2mg/kg IV (max 120mg)

Second choice:

Minor or major beta-lactam allergy
or history of carriage of MRSA
or during a ward outbreak of MRSA infection

Adults:

Single dose vancomycin 15mg/kg IV (max 2g)
plus gentamicin 3 mg/kg IV at induction

Paediatrics:

One month–18 years:

Single dose vancomycin plus gentamicin IV at induction (for dosage see below)

Vancomycin

One month-16 years

1 month – 6 months 10mg/kg

6 months – 1year 15mg/kg

1 – 12 years 17.5mg/kg

12-16 years 15mg/kg

Child 17-18 years

As adults

Gentamicin dosing:

One month-12 years

2.5mg/kg IV (max 120mg)

Child 12-18 years

2mg/kg IV (max 120mg)

Notes:

1. Aseptic, meticulous surgical technique is considered the most important factor in the prevention of post-operative infection. If hair is to be removed, use electric clippers with a single-use head on the day of surgery (NICE). Apply alcohol-containing solution to the operative site, avoiding pooling of solution in view of potential inflammable hazard, and wait at least 30 seconds before proceeding.
2. Shunts are irrigated in gentamicin before insertion. The efficacy in infection prevention is unproven.
3. Instil 20mg of [vancomycin](#) and 5mg of [gentamicin](#) into ventricles if an external ventricular drain is present before ventricular shunt implantation. Procedure for [administration of intraventricular and intrathecal antimicrobials via an external ventricular drain \(EVD\) or lumbar drain \(LD\)](#) and [Intrathecal monographs](#) are available on the intranet.
4. Drains must not be manipulated for any reason except clinical reasons and full aseptic precautions must always be followed. Drains must be removed as soon as possible to minimise the risk of meningitis.
5. Antibiotics may be continued for a maximum of 24 hours in prolonged and difficult operations or high risk procedures (eg elderly or those with malignancy).

5.2 Clean procedures in neuro-oncology patients

Neuro-oncology patients are at a high risk of infection including infections caused by *Propionibacterium* species. Please refer to section 5.3 below for prophylactic regimen.

5.3 Clean contaminated procedures and clean procedures in neuro-oncology patients

Clean contaminated procedures (including transmucosal, endoscopic and endonasal cranial procedures and surgery to acoustic neuromas).

Preparation of the mucous membrane is by irrigation with aqueous chlorhexidine.

First choice:

Adults:

Single dose co-amoxiclav 1.2 g IV at induction plus gentamicin 3 mg/kg IV at induction

Paediatrics:

One month–12 years:

**Single dose co-amoxiclav 30mg/kg IV
plus gentamicin 2.5mg/kg IV (max 120mg) at induction**

Child 12-18years:

**Single dose co-amoxiclav 30mg/kg (max 1.2 g) IV
plus gentamicin 2mg/kg IV (max 120mg) at induction**

Second choice:

**Minor or major beta-lactam allergy
or history of carriage of MRSA
or during a ward outbreak of MRSA infection**

Adults:

**Single dose vancomycin 15mg/kg IV (max 2g)
plus gentamicin 3 mg/kg IV
plus metronidazole 500 mg IV**

Paediatrics:

One month–12 years:

**Single dose vancomycin
1 month – 6 months 10mg/kg
6 months – 1year 15mg/kg
1 – 12 years 17.5mg/kg**

**plus gentamicin 2.5mg/kg IV (max 120mg)
plus metronidazole 7.5 mg/kg IV (max 500mg)**

12-18 years:

**Single dose vancomycin
12-16 years 15mg/kg
17-18 years as adults**

**plus gentamicin 2mg/kg (max 120mg)
plus metronidazole 7.5 mg/kg IV (max 500mg)**

5.4 Skull fracture and CSF fistula (including CSF rhinorrhoea and otorrhoea)

An urgent assessment is required and surgical repair may be indicated. Antibiotics are not recommended, except for neurosurgery, as above.

Adult patients should receive a single dose of pneumococcal polysaccharide vaccine (also known as PPV23 or Pneumovax II) 0.5ml IM.

Children above two years of age who completed the recommended routine pneumococcal conjugate vaccine (Prevenar-13) immunisation schedule, should receive a dose of PPV23 at least two months after the last dose of PCV13. For infants and children previously unvaccinated or partially vaccinated, refer to the [Green Book Chapter 25](#) for further information.

5.5 Prophylaxis of meningitis and brain abscess following penetrating craniocerebral injury

Duration of therapy is five days.

First choice:

Adults:

ceftriaxone 2g IV, once daily
Plus **metronidazole 500mg IV** eight hourly

This can be stepped down after three days of IV therapy to **co-amoxiclav 625mg PO** eight hourly for a further two days.

Paediatrics:

One month–12 years:

ceftriaxone IV once daily 50mg/kg IV (max 2g) once daily
plus **metronidazole 7.5 mg/kg IV (max 500mg) eight hourly**
(if the child is under 2 months old, metronidazole doses should be 12 hourly)

12-18 years:

ceftriaxone 2g IV daily
plus **metronidazole 7.5mg/kg IV (max 500mg) eight hourly**

Step-down after 3 days of IV therapy

One month–11 months:

co-amoxiclav 125/31 suspension 0.25ml/kg PO eight hourly

1–5 years:

**co-amoxiclav 125/31 suspension 0.25ml/kg PO eight hourly
or 5ml PO eight hourly**

6–11 years:

**co-amoxiclav 250/62 suspension 0.15ml/kg PO eight hourly
or 5ml PO eight hourly**

12-18 years:

co-amoxiclav 250/125 tablets PO eight hourly

Second choice (or major beta lactam allergy):

Adults:

**vancomycin 15mg/kg IV (max 2g) 12-hourly
plus ciprofloxacin 400mg IV 12-hourly
plus metronidazole 500mg IV eight-hourly**

Paediatrics:

One month–12 years:

vancomycin

1 month – 6 months 10mg/kg 6 hourly

6 months – 1year 15mg/kg 6 hourly

1 – 12 years 17.5mg/kg 6 hourly

plus ciprofloxacin 10mg/kg IV (max 400mg), eight-hourly

plus metronidazole 7.5 mg/kg IV (max 500mg), eight-hourly

(if the child is under two months old, metronidazole doses should be 12 hourly)

12-18 years:

vancomycin

12-16 years 15mg/kg 6 hourly

17-18 years as adults

plus ciprofloxacin 10mg/kg IV (max 400mg), eight-hourly

plus metronidazole 7.5mg/kg IV (max 500mg), eight-hourly

Notes:

Infection rates of penetrating injuries in the pre-antibiotic era were high with a mortality rate of around 50%. Reported infection rate with broad spectrum

antibiotics are between 1-5%. The risk of infection is increased with CSF leaks, wound sinuses and those crossing the ventricular system. Removal of foreign bodies and debridement of devitalised tissue is a vital component of therapy but is complicated by the possibility of damage to vital structures. Human anti-tetanus immunoglobulin (250-500 IU intramuscularly) should also be administered to all patients and their vaccination status be brought up to date. Refer to [tetanus immunisation, prophylaxis and treatment of tetanus prone wounds procedure](#) for more details. In the event of meningitis, brain abscess, wound infection or osteomyelitis, standard therapy should be instituted without delay, along with appropriate surgical exploration or excision.

5.6 Repeated doses post operatively and during long operations

The value of additional doses is not clearly defined. However national guidelines suggest that additional doses should be given at intervals of one to two times the half life of a drug so that adequate levels are maintained throughout the operation.

5.6.1 The following doses/ intervals are recommended for repeated dosing intra-operatively in adults

Co-amoxiclav 600mg IV four-hourly
Ciprofloxacin 400mg IV twelve-hourly
Flucloxacillin 1g IV four-hourly
Metronidazole 500mg IV six to eight-hourly
Vancomycin 500mg IV eight-hourly.

The following doses/ intervals are recommended for repeated dosing intra-operatively in paediatrics.

Paediatrics:

Flucloxacillin – 6 hourly

Child 1month – 18 years: 50mg/kg IV every six hours (max 2g every six hours)

Co-amoxiclav – 8 hourly (12 hourly if 1-3 months of age)

Child 1-3 months: 30mg/kg IV every 12 hours

Child 3 months-18 years: 30mg/kg (max 1.2g) every eight hours

Metronidazole – 8 hourly (but 12 hourly if 1-2 months of age)

Child 1 month-2 months: 7.5mg/kg IV every 12 hours

Child 2 months -18 years: 7.5mg/kg IV (max 500mg) every eight hours

Gentamicin – 8 hourly

One month–12 years: 2.5mg/kg IV (max 120mg)
Child 12-18 years: 2mg/kg IV (max 120mg)

Vancomycin

Child 1 month-18 years: adjusted according to plasma concentration
1 month – 6 months 10mg/kg 6 hourly
6 months – 1year 15mg/kg 6 hourly
1 – 12 years 17.5mg/kg 6 hourly
12-16 years 15mg/kg 6 hourly
17-18 years as adults

5.6.2 Antibiotics may be continued for a maximum of 24 hours after surgery in prolonged and difficult operations (eg complex spinal surgery)

Adults:

Ciprofloxacin 400mg IV twelve-hourly
Co-amoxiclav 1.2g IV eight-hourly
Flucloxacillin 1g IV six-hourly
Metronidazole 500mg IV eight-hourly
Vancomycin 15mg/kg IV (max 2g) twelve-hourly

Paediatrics:

Flucloxacillin

Child 1 month-18years 50mg/kg IV every six hours (max 2g every six hours)

Co-amoxiclav

Child 1–3 months 30 mg/kg IV every 12 hours
Child 3 months –18 years 30 mg/kg IV (max. 1.2 g) every eight hours

Metronidazole

Child 1 month -2 months 7.5mg/kg IV every 12 hours
Child 2months -18 years 7.5mg/kg IV (max 500mg) every eight hours

Vancomycin

Child 1 month - 18 years: adjusted according to plasma concentration

1 month – 6 months 10mg/kg 6 hourly

6 months – 1year 15mg/kg 6 hourly

1 – 12 years 17.5mg/kg 6 hourly

12-16 years 15mg/kg 6 hourly

17-18 years as adults

6 Monitoring compliance with and the effectiveness of the protocol

Prescribing of antibiotics is reviewed routinely for adherence to policy which offers the measurable standard in terms of recommended prescribing practices.

This review is achieved on daily ward rounds by medical teams and pharmacists, and more formally through clinical audit.

7 References

1. Bayston R, de Louvois J, Brown EM, Johnston RA, Lees P, Pople IK. Use of antibiotics in penetrating craniocerebral injuries. 'Infection in Neurosurgery' Working Party of British Society for Antimicrobial Chemotherapy. *Lancet*. 2000; **355**: 1813-7.
2. Brown EM, Pople IK, de Louvois J, Hedges A, Bayston R, Eisenstein SM, Lees P; British Society for Antimicrobial Chemotherapy Working Party on Neurosurgical Infections.
3. Spine update: prevention of postoperative infection in patients undergoing spinal surgery. *Spine*. 2004; **29**: 938-945.
4. Infection in Neurosurgery Working Party of the British Society for Antimicrobial Chemotherapy. Antimicrobial prophylaxis in neurosurgery and after head injury. *Lancet*. 1994; **344**:1547-51.
5. Barker F G. Efficacy of prophylactic antibiotics against meningitis after craniotomy: a meta-analysis. *Neurosurgery* 2007; **60**: 887-894.
6. Dellinger EP, Olsen KM, et al. Clinical practice guidelines for antimicrobial prophylaxis in surgery. *Am J Health-Syst Pharm*. 2013; **70**:195–283
7. De Jonge SW, Gans SL, Atema JJ et al. Timing of preoperative antibiotic prophylaxis in 54,552 patients and the risk of surgical site infection; A systematic review and meta-analysis. *Medicine*. 2017; **96**: 1-12
8. Summary of Product Characteristics are available at the link <https://www.medicines.org.uk/emc>

8 Associated documents

- [Central nervous system \(CNS\) infections guideline](#)
- [Administration of intraventricular and intrathecal antimicrobials via an external ventricular drain \(EVD\) or lumbar drain \(LD\)](#)
- [Intrathecal monographs](#)
- [Tetanus immunisation, prophylaxis and treatment of tetanus prone wounds guideline](#)

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Approval:	Mr M Garnett, Dr D Duane, Dr M Smith, August 2019		
JDTC approval:	n/a		
Owning department:	Neurosurgery		
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File name:	Neurosurgical prophylaxis Version6 August 2019.doc		
Supersedes:	Version 5, March 2017		
Version number:	6	Review date:	August 2022
Local reference:		Document ID:	8581